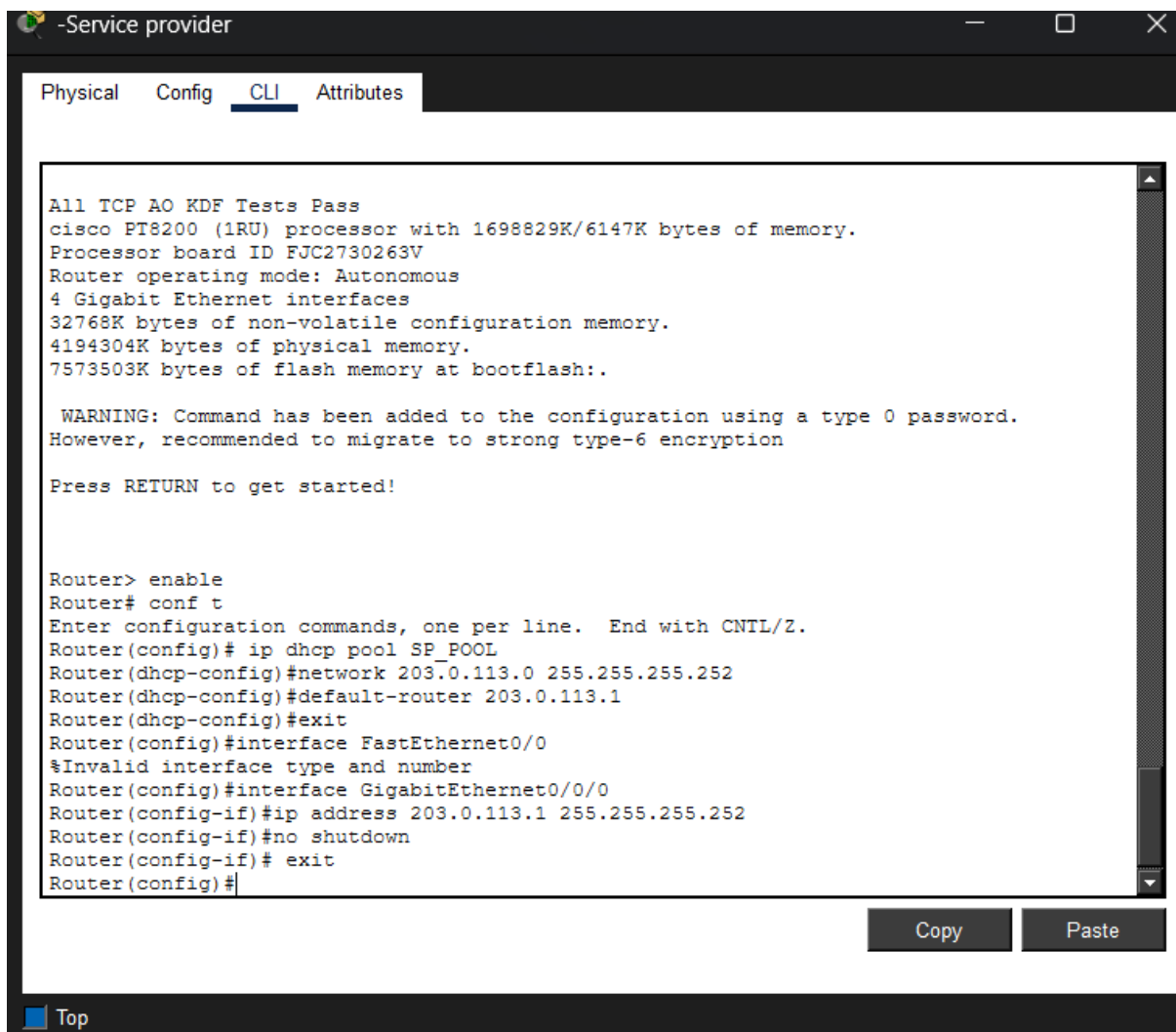


97.



The screenshot shows a terminal window titled "-Service provider" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the following text:

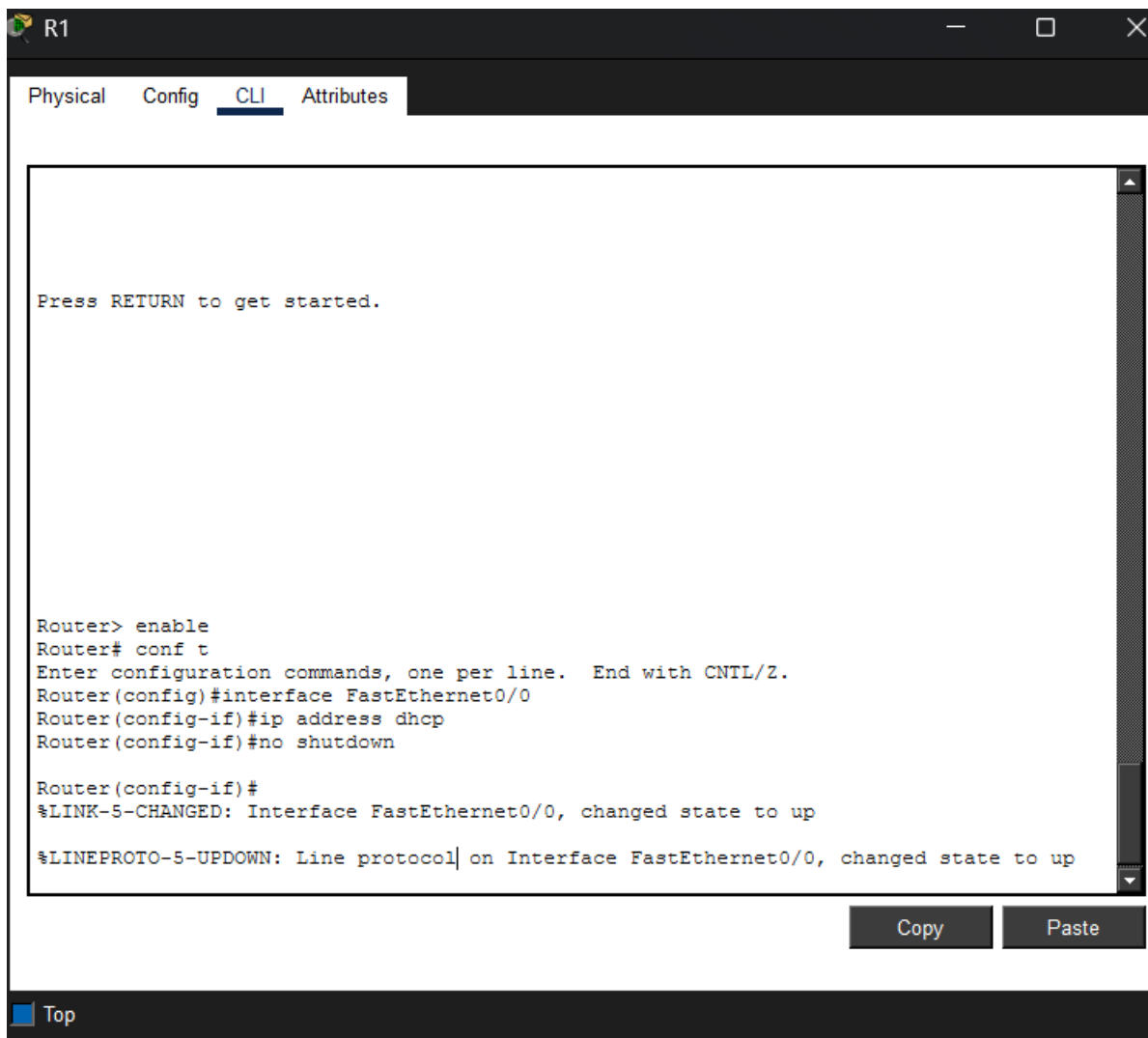
```
All TCP AO KDF Tests Pass
cisco PT8200 (1RU) processor with 1698829K/6147K bytes of memory.
Processor board ID FJC2730263V
Router operating mode: Autonomous
4 Gigabit Ethernet interfaces
32768K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
7573503K bytes of flash memory at bootflash:.

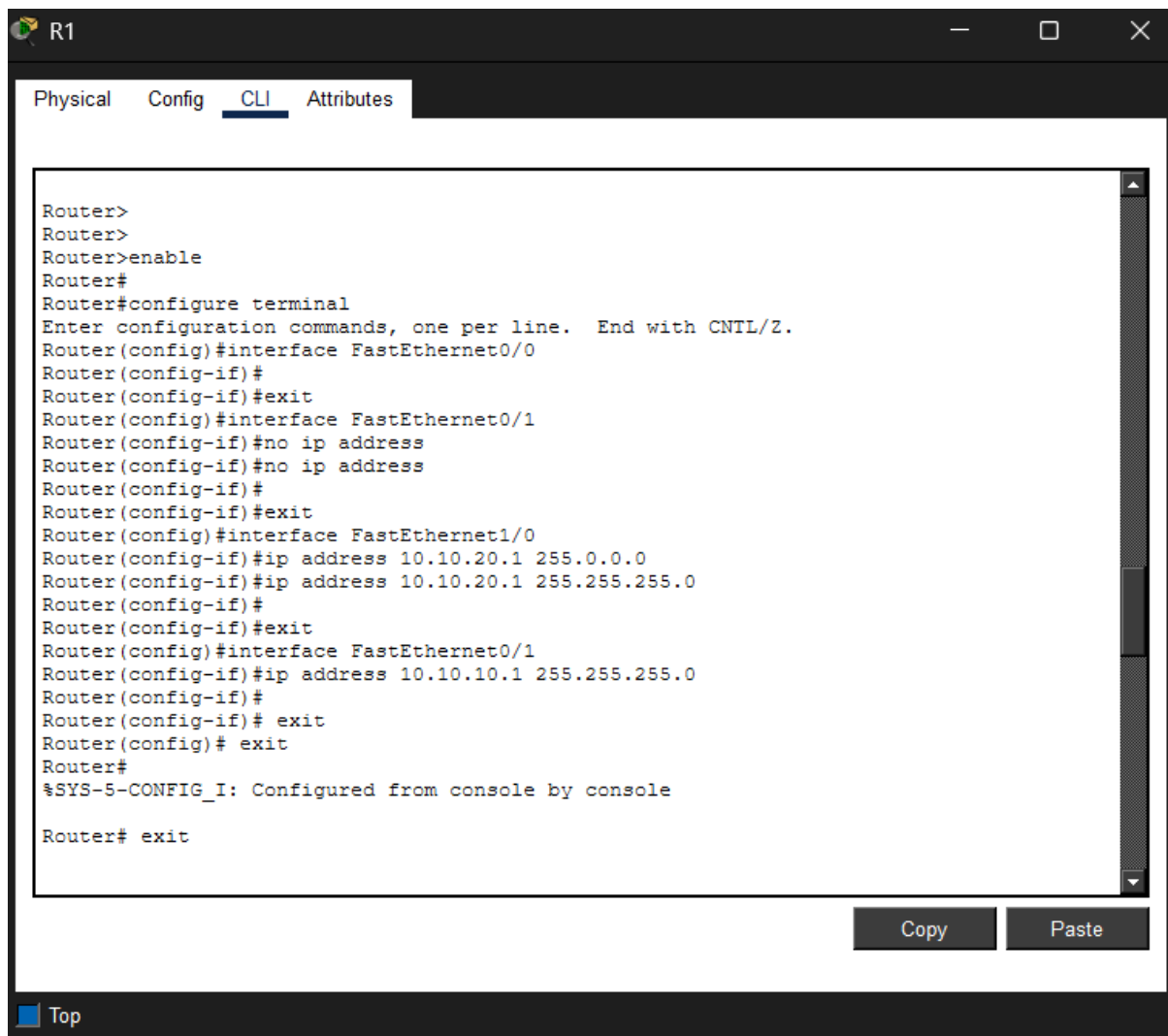
WARNING: Command has been added to the configuration using a type 0 password.
However, recommended to migrate to strong type-6 encryption

Press RETURN to get started!

Router> enable
Router# conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)# ip dhcp pool SP_POOL
Router(dhcp-config)#network 203.0.113.0 255.255.255.252
Router(dhcp-config)#default-router 203.0.113.1
Router(dhcp-config)#exit
Router(config)#interface FastEthernet0/0
%Invalid interface type and number
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 203.0.113.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)# exit
Router(config)#
```

At the bottom of the window, there are "Copy" and "Paste" buttons, and a "Top" link.



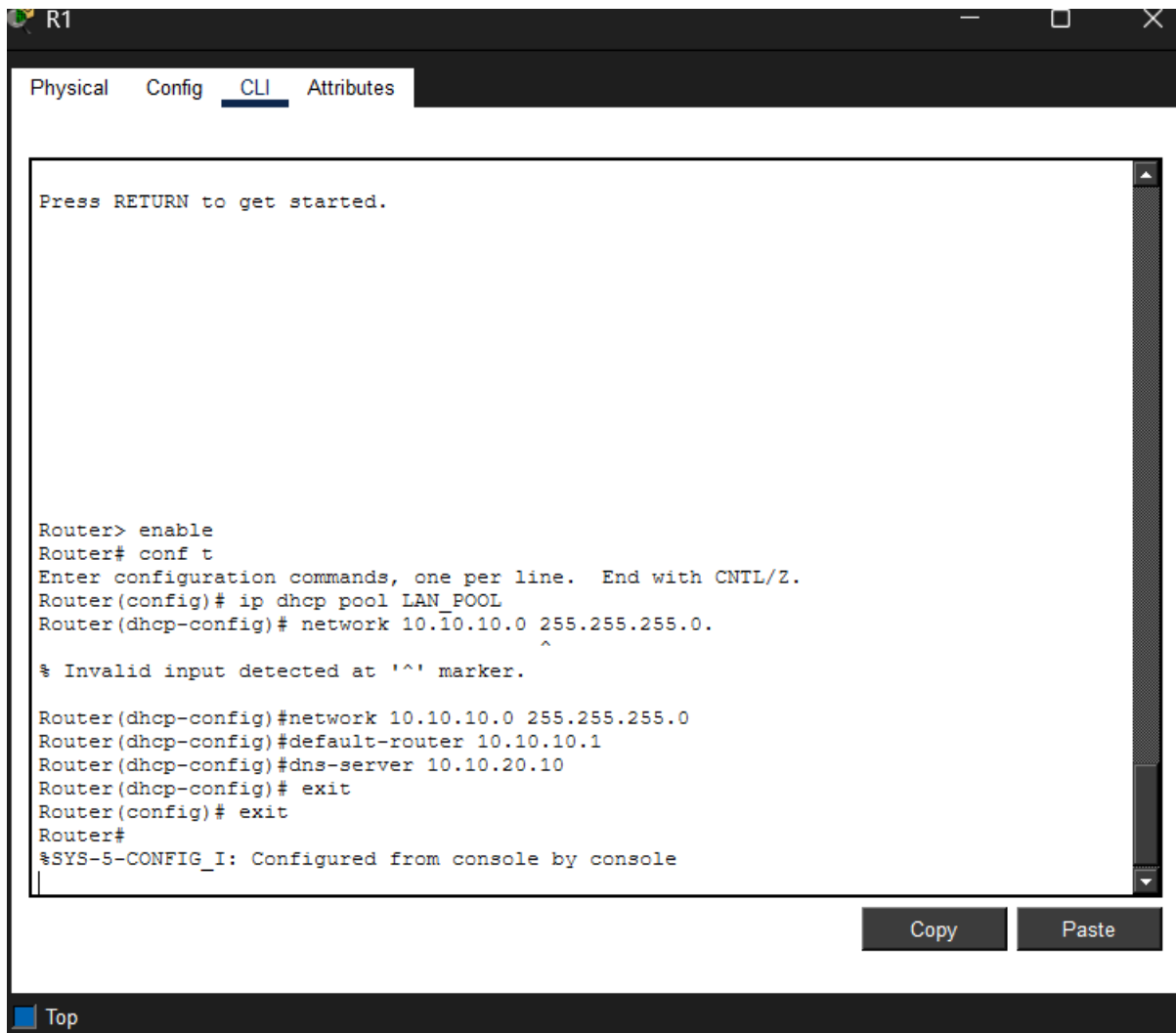


```
Router>
Router>
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#no ip address
Router(config-if)#no ip address
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet1/0
Router(config-if)#ip address 10.10.20.1 255.0.0.0
Router(config-if)#ip address 10.10.20.1 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 10.10.10.1 255.255.255.0
Router(config-if)#
Router(config-if)# exit
Router(config)# exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router# exit
```

Copy Paste

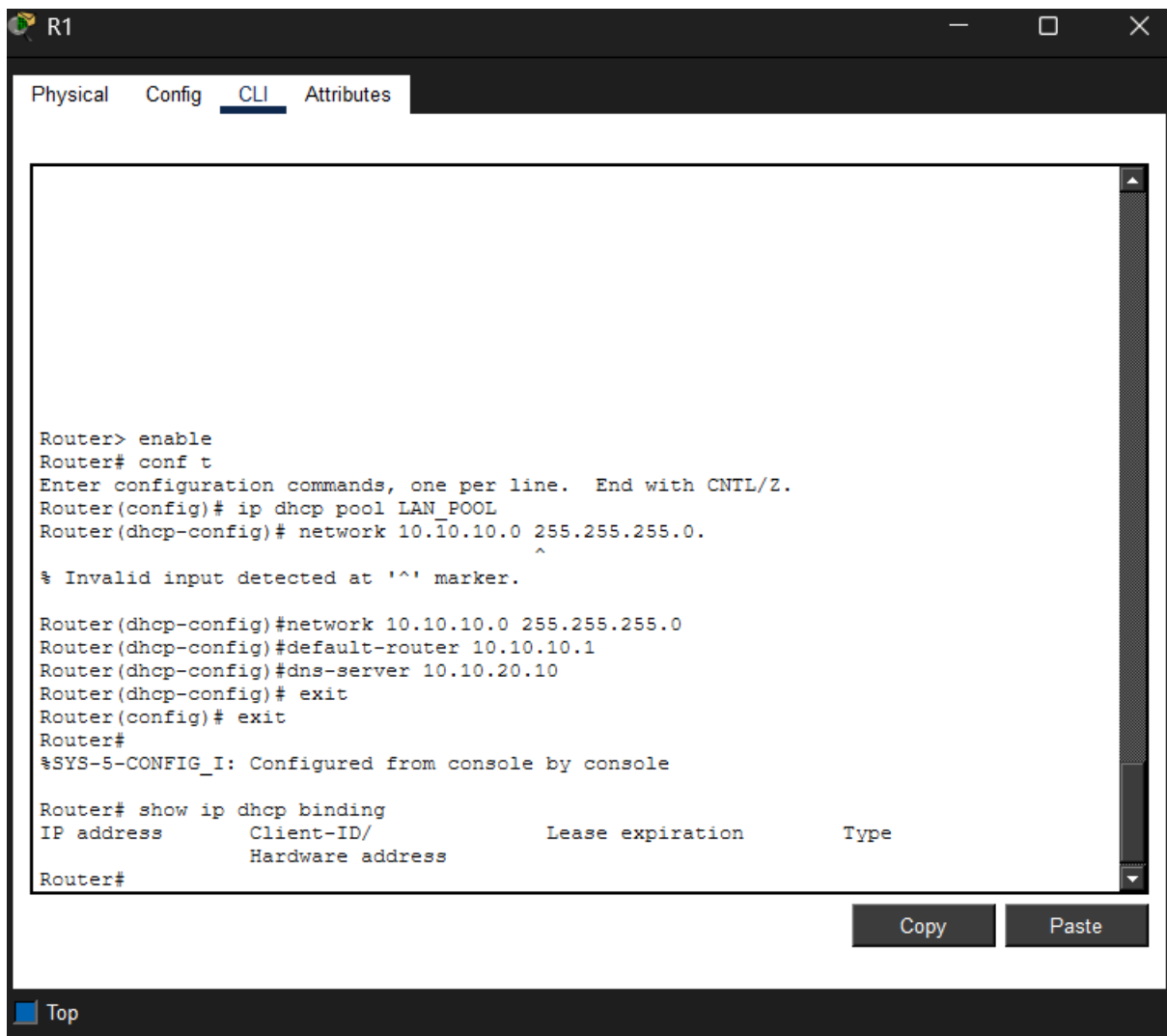
Top



The screenshot shows a terminal window titled 'R1' with tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, displaying a series of commands and their outputs. The text is as follows:

```
Press RETURN to get started.  
  
Router> enable  
Router# conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)# ip dhcp pool LAN_POOL  
Router(dhcp-config)# network 10.10.10.0 255.255.255.0.  
                                     ^  
% Invalid input detected at '^' marker.  
  
Router(dhcp-config)#network 10.10.10.0 255.255.255.0  
Router(dhcp-config)#default-router 10.10.10.1  
Router(dhcp-config)#dns-server 10.10.20.10  
Router(dhcp-config)# exit  
Router(config)# exit  
Router#  
%SYS-5-CONFIG_I: Configured from console by console  
|
```

At the bottom right of the terminal area, there are two buttons: 'Copy' and 'Paste'. At the bottom left of the window, there is a 'Top' button.



```
Router> enable
Router# conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# ip dhcp pool LAN_POOL
Router(dhcp-config)# network 10.10.10.0 255.255.255.0.
^
% Invalid input detected at '^' marker.

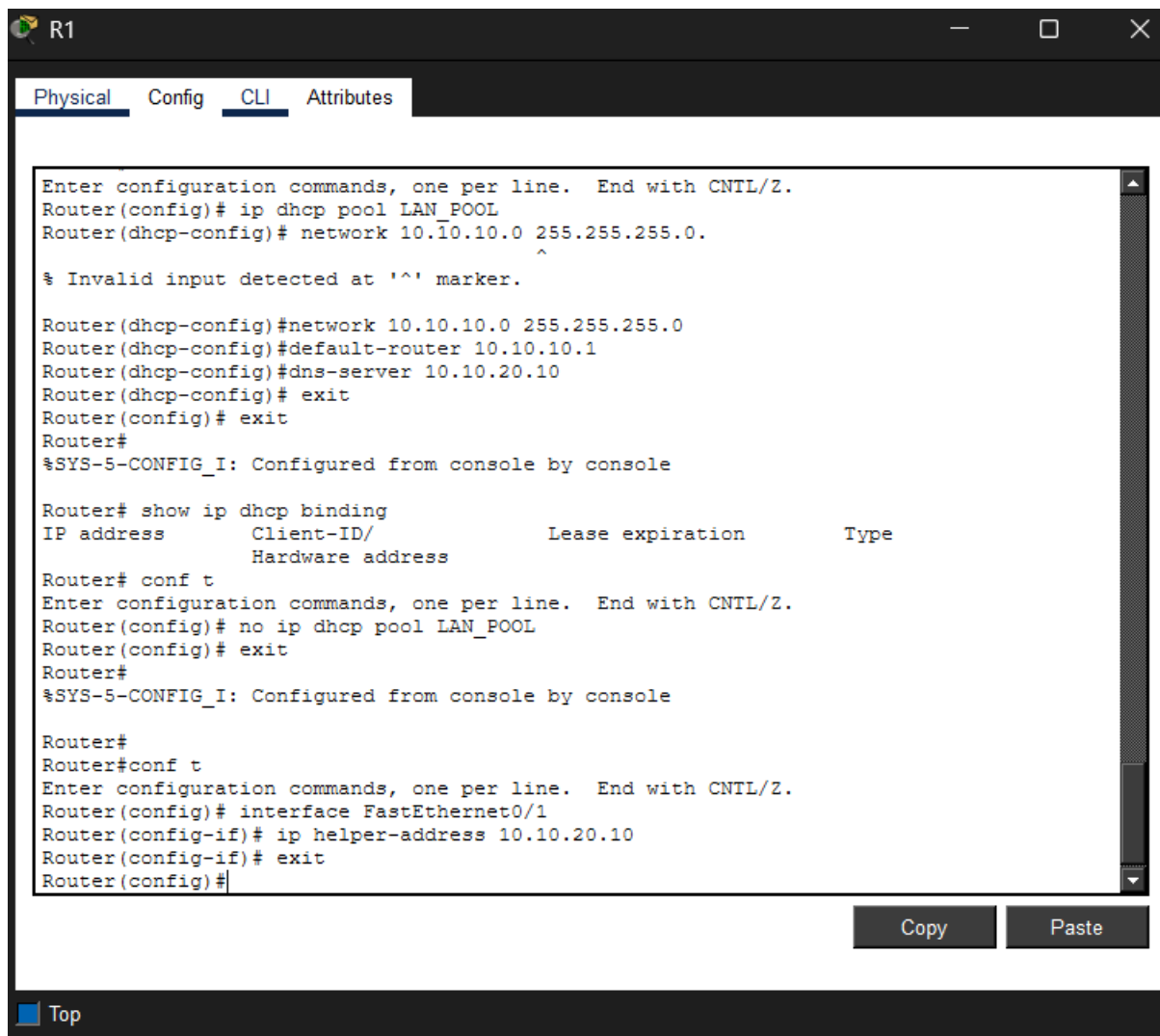
Router(dhcp-config)#network 10.10.10.0 255.255.255.0
Router(dhcp-config)#default-router 10.10.10.1
Router(dhcp-config)#dns-server 10.10.20.10
Router(dhcp-config)# exit
Router(config)# exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router# show ip dhcp binding
IP address      Client-ID/      Lease expiration        Type
                Hardware address
Router#
```

99.

Step 1:

Step 2:



The screenshot shows a Cisco R1 CLI window with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying a series of configuration commands and their outputs. The commands include creating a DHCP pool, setting the network, default-router, and dns-server, and then exiting the configuration mode. The output shows the configuration was successful. The window also has a 'Copy' and 'Paste' button at the bottom right and a 'Top' button at the bottom left.

```
R1
Physical Config CLI Attributes
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# ip dhcp pool LAN_POOL
Router(dhcp-config)# network 10.10.10.0 255.255.255.0.
^
% Invalid input detected at '^' marker.
Router(dhcp-config)#network 10.10.10.0 255.255.255.0
Router(dhcp-config)#default-router 10.10.10.1
Router(dhcp-config)#dns-server 10.10.20.10
Router(dhcp-config)# exit
Router(config)# exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router# show ip dhcp binding
IP address          Client-ID/          Lease expiration    Type
                   Hardware address
Router# conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# no ip dhcp pool LAN_POOL
Router(config)# exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# interface FastEthernet0/1
Router(config-if)# ip helper-address 10.10.20.10
Router(config-if)# exit
Router(config)#
```

Copy Paste

Top

Step 3:

DNS AND DHCP Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

FastEthernet0

serverPool

10.10.10.1

10.10.20.10

1010100

2552552550

156

0.0.0.0

0.0.0.0

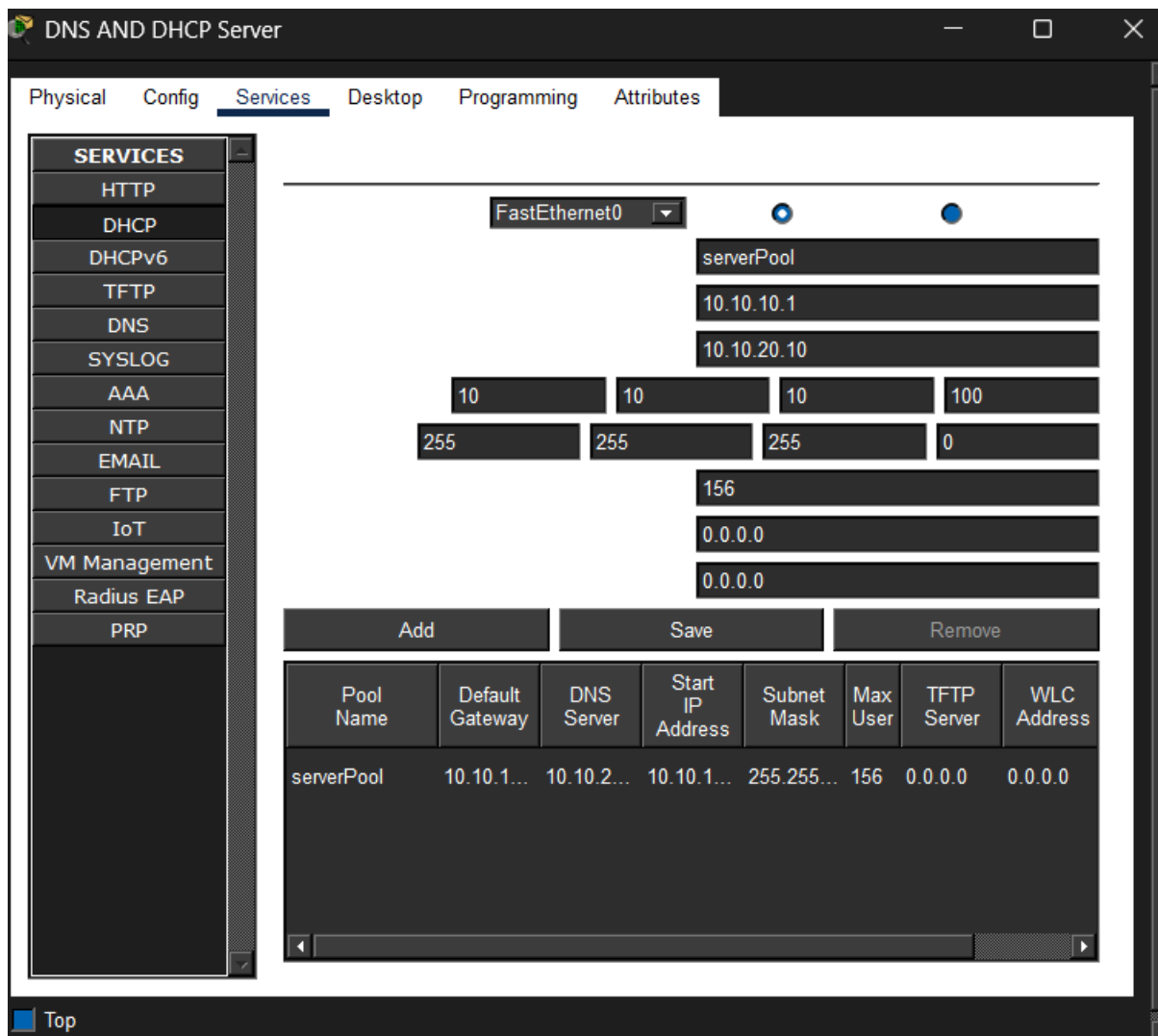
Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	10.10.1...	10.10.2...	10.10.1...	255.255...	156	0.0.0.0	0.0.0.0

Top



100.

DHCP client requests rely on Layer 2 broadcast message because the client does not yet have an IP address or know the network topology. Routers drop broadcast packets by default to prevent broadcast storms.

When the DHCP server sits on the same subnet, the broadcast reaches the server directly without crossing a router. When the server is on a different subnet, the router's interface drops the client's broadcast request. The IP helper address command configures the router to act as a DHCP Relay Agent. It intercepts the local Layer 2 broadcast, converts it into a Layer 3 unicast packet addressed directly to the DHCP server's IP address, and forwards it across the routed network.